

# Amber Harland-Bennett

 Oldham County, KY  
 [amberhb0108@hotmail.com](mailto:amberhb0108@hotmail.com) |  502-994-3477  
 | [Portfolio](#) | [GIS & Weather Portfolio](#)

## PROFESSIONAL SUMMARY

Detail-oriented GIS & Geospatial Data Analyst with hands-on experience in cartography, remote sensing, spatial analysis, and environmental hands-on research. Proven ability to process, analyze, and validate large geospatial datasets using industry-standard tools including ArcGIS Pro, ArcMap, ERDAS IMAGINE, and Excel. Experienced in both academic research and professional production environments, including maritime charting and satellite-derived environmental analysis. Passionate about climate, weather, and data-driven decision making.

## PROFESSIONAL EXPERIENCE

### Spatial Data Integrations (SDI) — *Remote*

Geospatial Analyst | March 2024 – August 2024

- Processed, edited, and quality-checked Electronic and Digital Nautical Charts (ENC/DNC) to ensure compliance with maritime safety standards
- Created, edited, and maintained shapefiles, geodatabases, and spatial features within production mapping workflows
- Utilized structured mapping tables to correct and normalize attribute data
- Executed feature extraction, attribution, spatial editing, and validation tasks to maintain high data accuracy
- Acquired, analyzed, compiled, and disseminated maritime safety information to support nautical database updates and chart production
- Collaborated within a remote production environment while meeting strict accuracy and quality benchmarks

### Murray State University — Department of Earth & Environmental Sciences

Student GIS & Research Assistant | February 2021 – May 2022

- Created raster maps using ArcMap with supporting data from USGS and Google Earth Pro
- Collected and recorded phenological data for 20 trees across campus for environmental monitoring
- Collaborated with a team to install and document an EOSFD Forced Diffusion sensor
- Processed and validated satellite-derived datasets using Excel
- Identified and corrected data inconsistencies and satellite errors
- Provided administrative and departmental support as needed

## **Kroger — Prospect, KY**

**Deli Associate** | September 2023 – December 2025

- Delivered consistent customer service in a high-volume retail environment
- Maintained food safety, sanitation, and operational standards
- Developed strong time-management, communication, and problem-solving skills

## **RESEARCH & PROJECT EXPERIENCE**

### **Above-Ground Biomass (AGB) Change Analysis — Global Scale**

- Analyzed global AGB trends, identifying an increase from 372 GtC (2010) to 728 GtC (2018)
- Utilized ArcGIS Pro, ERDAS IMAGINE 2020, Excel, and geoprocessing workflows
- Presented findings as a scholarly poster at Murray State University Scholars Week

### **Mayfield, KY Tornado Damage Analysis (December 10, 2021)**

- Gathered Sentinel-2 satellite imagery to assess tornado damage patterns
- Used ModelBuilder in ERDAS IMAGINE to calculate and visualize impact zones
- Produced thematic maps highlighting areas of highest damage concentration

### **Kriging Analysis of EF4–EF5 Tornadoes (1950–2019)**

- Analyzed historical tornado datasets to identify high-risk regions across Tennessee, Mississippi, Alabama, Oklahoma, and Kansas
- Applied spatial interpolation techniques to visualize probability trends

## **EDUCATION**

### **Murray State University**

Associate of Arts — General Studies | May 2023